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Sec: Star SC

Qualitative Analysis

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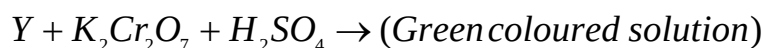
SINGLE ANSWER TYPE

- An aqueous solution of a given salt was made alkaline with solid NH_4Cl and NH_4OH solution. H_2S gas was then passed through the test tube. A black precipitate appeared. Which of the following conclusions is most appropriate about the precipitate. It could be
 - CuS or PbS
 - CoS or NiS
 - FeS or NiS or CoS
 - FeS or NiS or CoS or CuS or PbS
- Statement-I :** The blue precipitate formed by the action of $\text{K}_4[\text{Fe}(\text{CN})_6]$ on Fe^{3+} and by that of $\text{K}_3[\text{Fe}(\text{CN})_6]$ on Fe^{2+} have the same composition.
Statement-II : $[\text{Fe}(\text{CN})_6]^{3-}$ oxidises Fe^{2+} to Fe^{3+} and itself gets reduced to $[\text{Fe}(\text{CN})_6]^{4-}$ Correct relation is
 - both are correct statements and statement –ii is correct explanation for statement-i
 - both are correct statements and statement –ii is not correct explanation for statement-i
 - statement –I is correct but statement –II is false
 - statement –I is incorrect but statement – II is correct
- In which of the following all salts are almost insoluble in water?
 - LiF , AgF , Li_2CO_3
 - $\text{Fe}(\text{OH})_3$, PbCl_2 , MgF_2
 - CaCl_2
 - KCl , AgBr , ZnS
- A solid mixture of Hg_2Cl_2 , $\text{K}_2\text{Cr}_2\text{O}_7$ is heated with conc. H_2SO_4 and it produces
 - greenish-yellow gas
 - Colourless gas
 - red coloured gas
 - no gas
- A metal nitrate reacts with KI to give a brown precipitate which on addition of excess of KI convert into orange colour solution. The cation of metal nitrate is:
 - Hg^{2+}
 - Bi^{3+}
 - Pb^{2+}
 - Cu^+
- An aqueous solution of a substance gives a white precipitate on treatment with dilute hydrochloric acid, which dissolves on heating. When hydrogen sulphide is passed through the hot acidic solution, a black precipitate is obtained. The substance is a :
 - mercurous salt
 - Cr^{2+} salt
 - Ag^+ salt
 - Pb^{2+} salt
- Read of the following statements and choose the correct code w.r.t true(T) and false(F).
 - manganese salts give a violet borax bead test in reducing flame
 - from a mixed precipitate of AgCl and AgI , ammonia solution dissolves only AgCl
 - ferric ions give a deep green precipitate, on adding potassium ferrocyanide solution
 - on boiling the solution having K^+ , Ca^{2+} and HCO_3^- we get a precipitate of $\text{K}_2\text{Ca}(\text{CO}_3)_2$
 - TTFF
 - FTFT
 - FTFF
 - TTFT

8. Which one of the following metal sulphides has maximum solubility in water?
 a) HgS b) CdS c) SrS d) ZnS
9. What happens when ammonium thiocyanate is added to the aqueous solution of CuSO_4 followed by passing SO_2 through the solution.
 a) First black ppt appears, which reduced to white ppt
 b) first white ppt appears, which reduced to black ppt
 c) first black ppt appears, which on reduction also appears as black ppt
 d) first white ppt appears, which on reduction also appears as white ppt.
10. Which of the following pair of salts forms precipitate with excess of NH_4OH
 a) $\text{CuSO}_4, \text{CdSO}_4$ b) $\text{Cu}(\text{NO}_3)_2, \text{ZnSO}_4$ c) $\text{FeCl}_3, \text{AlCl}_3$ d) all the above

MULTIPLE ANSWER TYPE

11. AgCl is soluble in
 a) water b) $\text{NH}_3(\text{aq})$ c) KCN d) dil HCl
12. A yellow precipitate is obtained when
 a) A lead acetate solution is treated with K_2CrO_4
 b) A $\text{Pb}(\text{NO}_3)_2$ solution is treated with KI
 c) An AgNO_3 solution treated with KI
 d) H_2S is passed through a solution of CdSO_4
13. A metal nitrate on reaction with excess sodium hydroxide solution gives a white precipitate, but it rapidly turns to brown on adding bromine water. Aqueous solution of metal nitrate produces a pink precipitate with disodium hydrogen phosphate solution in the presence ammonia. The cation is
 a) Cu^{2+} b) Mn^{2+} c) Mg^{2+} d) Zn^{2+}
14. A black mineral (A) in solid state is fused with KOH and KNO_3 and the mixture extracted with water to get a green coloured solution (B). On passing CO_2 gas through the solution the colour changes to pink with a black residue (C)
 Which of the following is/are correct
 a) The pink colouration is decolourised by acidified ferrous sulphate solution
 b) The black residue is same as compound (A)
 c) When O_3 is passed through the green colour solution is changes to pink.
 d) Shape of the anion of pink substance is tetrahedral
15. $\text{A}(\text{Black}) + \text{dil } \text{H}_2\text{SO}_4 \rightarrow \text{B}(\text{aq}) + \text{C}(\text{gas})$
 $\text{C} + (\text{CH}_3\text{COO})_2\text{Pb} \rightarrow \text{Black}$
 $\text{B} + \text{NaOH} \rightarrow \text{green ppt}(\text{D})$ changing to brown (E) by atmospheric O_2
 $\text{B} + \text{K}_3[\text{Fe}(\text{CN})_6] \rightarrow \text{Blue}(\text{F})$
 $\text{E} + \text{HCl} \xrightarrow{\Delta} \text{deep yellow solution}(\text{G})$ $\text{F} + \text{KCNS} \rightarrow \text{Red coloured solution}(\text{H})$
 Correct statements about this are / is
 a) Green ppt is insoluble in excess of NaOH
 b) Deep yellow compound (G) can give red/reddish brown ppt with alkali
 c) Blue coloured F has both +2 and +3 oxidation states for metal
 d) Metal in A is iron
16. From the following information,
 $\text{X} + \text{H}_2\text{SO}_4 \rightarrow \text{Y}(\text{a colourless and irritating gas})$



Identify the pair X and Y from the list given below:

- a) Cl^- , HCl b) S^{2-} , H_2S c) CO_3^{2-} , CO_2 d) SO_3^{2-} , SO_2

17. A white crystalline salt A reacts with dilute HCl to liberate a suffocating gas B and also forms a yellow precipitate. The gas turns potassium dichromate acidified with dilute H_2SO_4 to a green coloured solution C. A, B and C are respectively:
 a) Na_2SO_3 , SO_2 , $Cr_2(SO_4)_3$ b) $Na_2S_2O_3$, SO_2 , $Cr_2(SO_4)_3$
 c) Na_2S , SO_2 , $Cr_2(SO_4)_3$ d) Na_2SO_4 , SO_2 , $Cr_2(SO_4)_3$
18. Which of the following reagents cannot be useful in separating a mixture of Zn^{+2} and Cu^{+2}
 a) H_2S in acidic medium b) H_2S in alkali medium
 c) $NaOH$ solution d) NH_3 Solution
19. Which of the following reagents can be used to distinguish between SO_2 and CO_2
 a) Lime water b) $BaCl_2$ c) $H_2O_2 + BaCl_2$ d) Acidified dichromate paper
20. A solution of a metal ion, when treated with KI , gives a red precipitate which dissolves in an excess of KI to give a colourless solution. However the solution of the metal ion on treatment with a solution of cobalt(II) thiocyanate gives rise to a deep blue crystalline precipitate. The metal ion is
 a) Pb^{+2} b) Cu^{+2} c) Hg^{+2} d) Co^{+2}
21. Which of the following will dissolve in a mixture of $NaOH$, H_2O_2
 a) $Fe(OH)_3$ b) $Cr(OH)_3$ c) $Al(OH)_3$ d) $Zn(OH)_2$
22. The pair of ions can be separated by using dil HCl
 a) Ag^+ and Cu^{+2} b) Ag^+ and Hg_2^{2+} c) Hg_2^{2+} and Cd^{+2} d) Ag^+ and Al^{+3}
23. Identify the compound which does not respond to chromyl chloride test?
 a) NH_4Cl b) KCl c) $HgCl_2$ d) $SnCl_4$

INTEGER TYPE

24. How many of the following are black coloured compounds
 Ag_2S , PbS , HgS , CuS , FeS , CoS , NiS , ZnS , CdS , Sb_2S_3 , MnS
25. Dimethyl glyoxamate ion with Ni^{+2} forms rosy red colour in alkaline medium. The coordination number of metal ion in complex is
26. Residue $Mn(OH)_2$ is added with few drops of conc. Nitric acid followed by a compound to get purple colour. This is confirmation of Mn. the compound can be how many of the following Na_3BiO_4 , PbO_2 , Pb_3O_4 , SnO , Bi_2O_3 , $NaIO_4$, $(NH_4)_2S_2O_8$
27. On adding excess of chlorine water to potassium iodide solution containing little carbon tetra chloride gives first violet colour to carbon tetrachloride and then violet colour is discharged. The change in the oxidation state of iodine from initial to final compound is
28. How many of the following metal sulphides are soluble in hot dilute nitric acid PbS , HgS , Ag_2S , CuS , CdS , Bi_2S_3 , NiS , CoS

PASSAGE-1

A white solid A reacts with dilute H_2SO_4 to produce a colourless gas B and a colourless solution C. The reaction between B and acidified dichromate yields a green solution and a slightly coloured precipitate D. The substance D, when burnt in air, gives a gas E which reacts with B to yield D and a colourless liquids. Anhydrous copper sulphate turns blue with this colourless liquid. The addition of aqueous NH_3 or $NaOH$

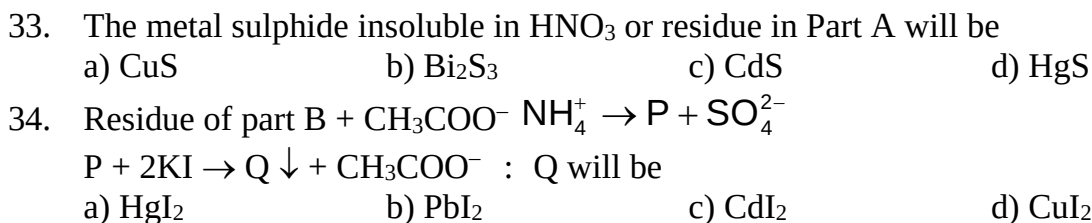
Answer the following questions based on above passage.

- ### PASSAGE-2

31. The compound (A) is
a) ZnCO_3 b) PbCO_3 c) Na_2CO_3 d) $(\text{NH}_4)_2\text{CO}_3$

32. The compound (C) is
a) PbO b) ZnO c) Na_2O d) Ag_2O

PASSAGE-3



KEY

[illegible]